Engineering Mechanics 1 · Unit_01 · UNIT 1

DRAFT — UE26ME142A · Unit_01 · UNIT 1 · UE26ME142A Unit 1 Class 11 AVSummary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 1 Class 11 AVSummary

1 UE26ME142A Unit 1 Class 11 AVSummary

Study focus: UE26ME142A_Unit 1_Class 11_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000234 · EV0000235

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Dr.

Key details

- Svinkitesh from Civil Engineering under Department of Science and Humanities BS University.
- In the last class, we had discussed about the solving of problems on couples, and I had also informed you that in the next class, that is class number fourteen, I would be starting the last topic of unit one, that is resultant.
- What is resultant?
- Resultant, I had also mentioned you regarding resultant in the very few first classes.
- Resultant is that single force which would produce the same effect as that of us system of forces.
- If you apply the system of forces on a physique body, there could be some effect.
- The effect could be a translatory motion or a rotatory motion or the combination of these two.
- It can be translatory as well as rotatory.
- This single force would also produce the same effect as that of that system of forces, then this single force would be called as the resultant.
- I talked about dot product and when I talked about dot product I told you if you have vectors or if you represent the forces as vectors and get the dot product of this vectors, that would give you the resultant.
- That means it would give you the vector form of the resultant.
- That means it will give you the magnitude of the component of the resultant along x axis, y axis.

- It would also give you the sense of the components of the resultant along X along y.
- Then it would also tell you what is the direction of these components, that is, it is along X and along Y.
- You should also take the sign of the forces or the vector into consideration.
- This f sigma f x will be in the form of Fx i or RX i plus R y j or Rx I minus R y j.
- What about the sign convention according to the forces that is acting.
- The sum of all the forces or the component of all the forces acting along X, algebraic sum of all the components of the forces acting along X, which should be nothing but your Rx.
- Components of the forces acting along y, which will be r y, and then I can find out the magnitude of the resultant R is equal to root of summation of R square plus summation of R y square.
- Would give us the magnitude of the resultant.
- The theta will give us the direction of the resultant with respect to the x-axis.
- Based on whether this is positive or negative, I would get the sense of the component of the resultant along x axis positive or negative, or y axis positive or negative.
- When we club these two, I get the sense of the resultant.
- The same thing is explained here.
- Say if you have a rigid body.
- And there are four three forces acting F1, F2, and F3.
- Now, because of these two forces F2 and F3, I can find out the resultant as R1 using the law of parallelogram of forces.

Example or application from the lecture

- Let us look at this.
- For example, you have a system of forces.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000234 · EV0000235 · EV0000236 · +1 source units in pack.json

Core explanation · 05:00–09:59

Adjacent sides of the parallel and then find out the resultant R.

Key details

- This is one way of finding out the resultant.
- The other way of finding out the resultant is explained here.
- Each force you find out the component of the forces.
- X and Y similarly for F2 again F2x and F2Y for F3 also again you find out F3x and F3Y.
- F3x is also positive, whereas F1 y is positive.
- F2y is also positive, whereas F3Y is negative.
- Adding all the components.
- Algebraically I get R y.
- We know that for a force there will be four characteristics.
- What are these four characteristics?
- I had explained this earlier also, I will repeat it.

- First one is the magnitude of the force.
- Or the quantity of the force is the magnitude.
- The second one is the direction, which is the angle made by the force with respect to a reference axis.
- The sense or line of action of the force, whether the force is moving away from the point of application or moving towards the point of application, and the fourth one is the point of application.
- If all the three forces have the same point of application, or if it is a.
- Concurrent force system, the resultant will also have the same point of application.
- But if it is not a concurrent force system, as you can see here, this is one force, this is another force, this is another force.
- When we produce these three forces, all the three of them will not meet at the same point.
- In such case, where am I supposed to apply the resultant force is the question.
- To find out the point of application of the resultant, we go for what is called as principle of movement, or it is also called as extension of varignos theorem.
- Or it is also called as law of movements.
- I would define all this.
- If you remember I defined this when I was talking about the second topic in unit one, chapter two, that is movement.
- What does Varing North theorem state?
- Varing North theorem states that if you have a force, the moment produced by the force would be equal to the moment produced by its force components about the same point.
- Very importantly, about the same point is very important.
- The moment produced by a force.
- About a point is equal to the moment produced by the component of the force about the same point.
- Now coming to principle of moments or law of moments or extension of Varingon's theorem means.
- The moment produced by a system of forces about any point will be the same as the moment produced by their resultant about the same point.
- I'll repeat.
- The moment produced by a system of forces about any point.
- Is equal to the moment produced by their resultant about the same point.

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EV0000238 · EV0000239 · EV0000240 · +1 source units in pack.json

Core explanation · 10:00–14:59

And they have mentioned the distances D1, D2, and D3, which are the perpendicular distance of this force F1 and F2 and F3 from point O.

Key details

- That means this angle is 90 degrees, this angle is also 90 degrees, this angle is also 90 degrees.

- Along with this, I will also apply this moment M1, M2, and M3, which is nothing but the force couple system.
- If you remember, so instead of applying this force at this point A, F1, F2, and F3, I can apply these forces at O by a force couple system, and that's what is indicated here.
- The sense of that.
- If I take this about this point the moment will be anti-clockwise and that is what is indicated here.
- Then if I take this force, the moment about this point will be this force into this distance, which will again be clockwise, and that is what is indicated here as M3.
- That is equal to summation F into D.
- Then you add all these three, that is dot product, you get R, which is sigma F.
- Somewhere away from O at a distance perpendicular to this R, that is D.
- The four characteristics.
- One is the magnitude, which is the dot product of all the forces acting on the body.
- Then how do we get the direction of the resultant?
- We can either find out Rx and R y.
- And then find out theta which is tan inverse of R y by R x where theta is the angle made by the resultant with the x axis.
- To find out the sense of the resultant, you have Rx and R Y, merge these two, whatever you are going to get will be the sense of the resultant.
- Then to determine the.
- Point of application of the resultant we should go for principle of movement or what we call it as the extension of variaus or law of movements.
- There could be a situation where we get only a moment as a resultant.
- We may not get the force as a resultant.
- I would take F1 and F2 and find out the resultant of these two, which would be this.
- Which would be this.
- That is, I just have completed the parallelogram.
- The diagon passing through the intersection of the two forces will give the resultant.
- This is the resultant R.
- But if this resultant that we get, okay, if that resultant of F1 and F2 that we are going to get as is will be equal to the magnitude of F3, which is acting as an external force on the rigid body.
- But in the opposite sense, as indicated here as minus F3, this would be the resultant of F1 and F2, which would be the same as the magnitude of F3, but the sense would be opposite.
- Then now you can see that these two.
- forms a couple and we know that the characteristics of a couple is the characteristics of a couple is to make the body rotate only and we also know that one of the.

Example or application from the lecture

- To find out what is the movement of all these three forces about this point O, which is given here.
- Okay now for that I have taken this example.
- Say for example we have F1, F2, and F3, three forces acting on a body.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000242 · EV0000243 · EV0000244 · +1 source units in pack.json

Core explanation · 15:00–19:59

characteristics of a couple S the net force acting because of a couple S zero.

Key details

- Like this, the other one is acting like this.
- The magnitude of that would be equal to F3 into D.
- F3 into D.
- So because now these two when I take it the moment produced will be a clockwise moment of magnitude which will be equal to F3 into D.
- And we also know about which axis it is acting, it is acting about the z axis which is coming out of the plane.
- No, where should we apply?
- We have already discussed that moment is a.
- Mom is a free vector.
- You can apply the moment anywhere you want.
- That would be the resultant of these three forces.
- Which is going to be acting.
- The resultant of these three forces now is just a moment of magnitude F3 into D, which would be acting in the clockwise direction along the z axis.
- You can apply it anywhere you want.
- The law of transmissibility of forces will apply, or the principle of transmissibility of forces will apply.
- The point of application of the vector can be anywhere, provided it is in contact with the body.
- Then only the body will experience that.
- I hope it is clear to everybody.
- In today's class, we have discussed the last topic of unit one, which is under chapter two.
- That is resulant.
- Resultant is that single force which produces the same effect as that of a system of forces.
- How do we find the resultant?
- The magnitude of the resultant is nothing but the dot product of all the forces.
- The direction is nothing but tan inverse of R y by Rx, provided the direction is with respect to the x axis.
- Where R is the component of all the forces along Y, and Rx which is the component of all the forces along X, R it is the component of the resultant about X and Y.
- Then we also have seen how to get the sense, you merge the Rx and R y together, you get the sense of the resultant R.
- Then to determine the point of application of the single force, you have to go for extension of Arianov's principle or principle of moments or law of moments, which states that the moment produced by a system of forces is equal to the moment produced by the.
- Resultant about the same point.

- And we have also looked at what could be the result.
- Resultant could be that only one single force or it could be a movement also.
- That what how do we get only one single movement is what I have explained in this last slide in this class.
- We will be continuing our class in the next class, that is class number 15, where I will take up and solve few problems related to resultant.
- We will meet in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000246 · EV0000247 · EV0000248 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 1_Class 11_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/011_UE26ME142A_Unit_1_Class_11_AVSummary__3d8f98864b87c9b2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 11_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/011_UE26ME142A_Unit_1_Class_11_AVSummary__3d8f98864b87c9b2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 11_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/011_UE26ME142A_Unit_1_Class_11_AVSummary__3d8f98864b87c9b2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 11_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/011_UE26ME142A_Unit_1_Class_11_AVSummary__3d8f98864b87c9b2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md have no readable `.slides.md` text; their original page images are included.
4. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md have no readable `.slides.md` text; their original page images are included.
5. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md have no readable `.slides.md` text; their original page images are included.
7. 3 rendered pages in 004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md have no readable `.slides.md` text; their original page images are included.
8. 3 rendered pages in 005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md have no readable `.slides.md` text; their original page images are included.
9. 2 rendered pages in 006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md have no readable `.slides.md` text; their original page images are included.
10. 3 rendered pages in 008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md have no readable `.slides.md` text; their original page images are included.

11. 2 rendered pages in 009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md have no readable `.slides.md` text; their original page images are included.
12. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md have no readable `.slides.md` text; their original page images are included.
14. 2 rendered pages in 011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md have no readable `.slides.md` text; their original page images are included.
16. 4 rendered pages in 012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md have no readable `.slides.md` text; their original page images are included.
17. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
18. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
19. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
20. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
21. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00054	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0055. Sidecar variants compare...
SRC00055	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00056	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0055. Sidecar variants compare...
SRC00057	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0055. Sidecar variants compare...

C Revision and retrieval practice